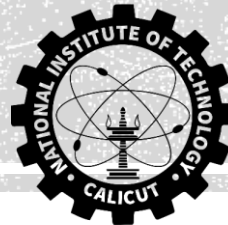


QUALITY, RELIABILITY AND MAINTENANCE ME327E



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Operating Characteristic Curve

- In practice, however, the shape of the OC curve is not ideal.
- If the observed number of nonconforming items or nonconformities is less than or equal to c , the lot is accepted. Otherwise, the lot is rejected.
- To construct a type A OC curve, we assume that the sample is chosen from an isolated lot of finite size.
- The probability of accepting the lot is calculated based on a **hypergeometric distribution**.
- The probability of finding x nonconforming items in the sample is given by, $P(x) = \frac{\binom{D}{x} \binom{N-D}{n-x}}{\binom{N}{n}}$

Where D represents the number of nonconforming items in the lot. Since the lot will be accepted if c or fewer nonconforming items are found, the probability of lot acceptance is

$$P_a = P(x \leq c) = \sum_{x=0}^c P(x)$$

Operating Characteristic Curve

- To construct a type B OC curve, we assume that a stream of lots is produced by the process and that the lot size is large (atleast 10 times) compared to the sample size.
- A binomial distribution can be used to find the probability of observing x nonconforming items in a sample of size n .
- Assuming the lot proportion nonconforming is p , this probability is given by,
$$P(x) = \binom{n}{x} p^x (1 - p)^{n-x}$$
- If the lot size is large and the probability of a nonconforming item is small
- Poisson distribution can be used as an approximation to the binomial distribution.
- The probability of x nonconforming items in the sample is found from,
$$P(x) = \frac{\lambda^x e^{-\lambda}}{x!}$$

Where, $\lambda = np$ represents the average number of nonconforming items in the sample

Exercise

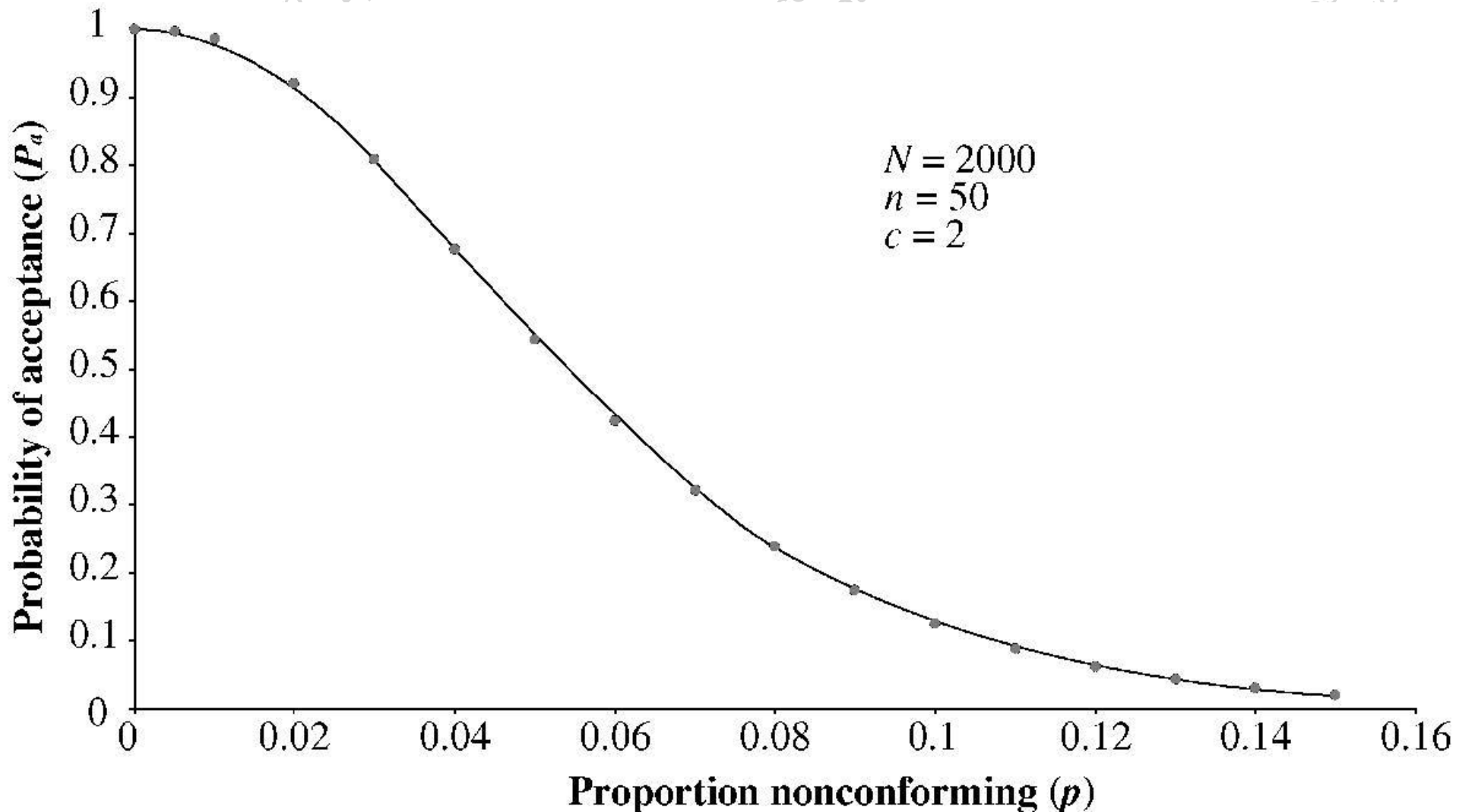


Construct an OC curve for a single sampling plan where the lot size is 2000, the sample size is 50, and the acceptance number is 2.

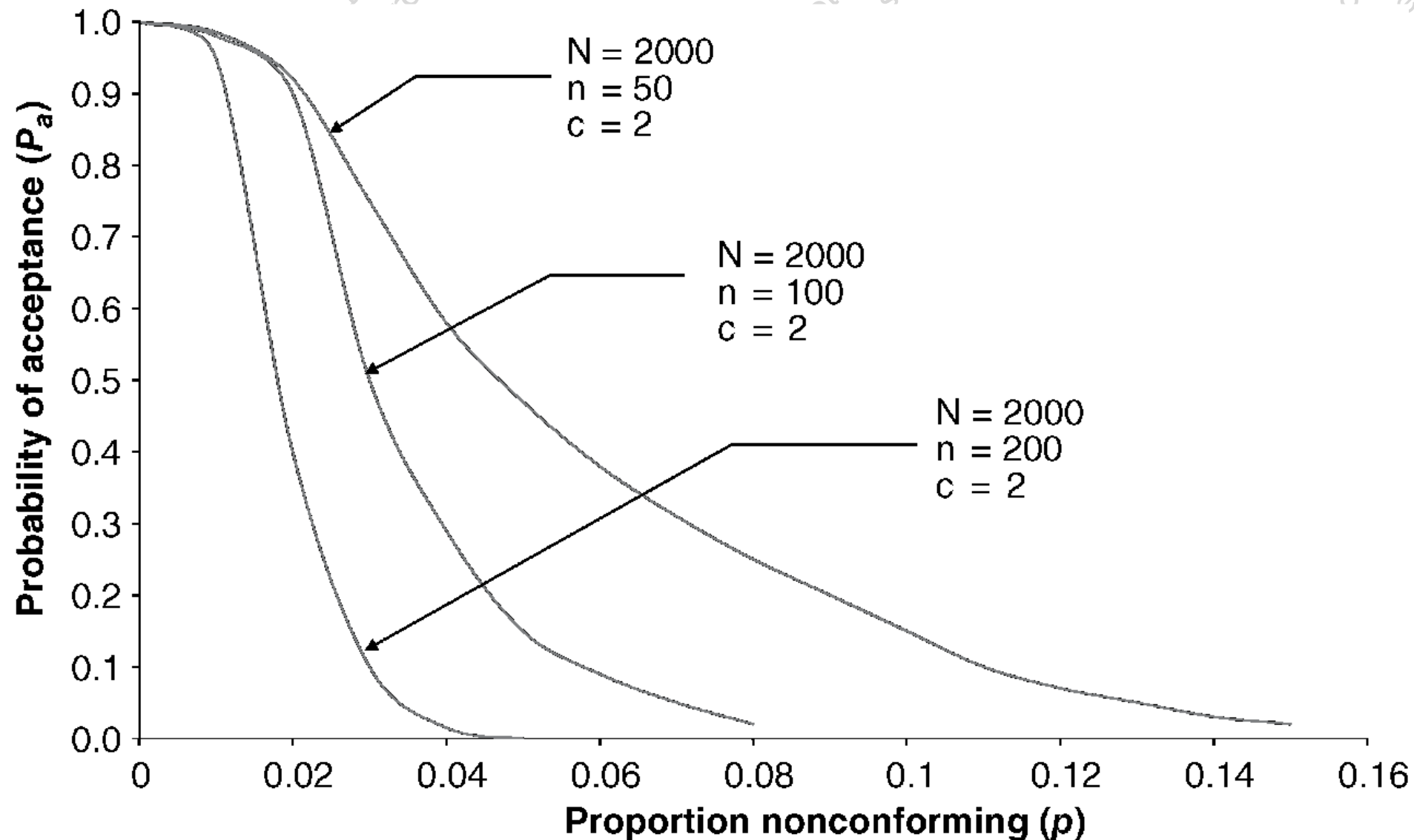
Lot Acceptance Probabilities for Different Values of Proportion Nonconforming for the Sampling Plan $N = 2000, n = 50, c = 2$

Proportion Nonconforming,		Probability of Lot Acceptance,	Proportion Nonconforming,		Probability of Lot Acceptance,
p	np	P_a	p	np	P_a
0.0	0.0	1.000	0.08	4.00	0.238
0.005	0.25	0.997	0.09	4.50	0.174
0.01	0.50	0.986	0.10	5.00	0.125
0.02	1.00	0.920	0.11	5.50	0.088
0.03	1.50	0.809	0.12	6.00	0.062
0.04	2.00	0.677	0.13	6.50	0.043
0.05	2.50	0.544	0.14	7.00	0.030
0.06	3.00	0.423	0.15	7.50	0.020
0.07	3.50	0.321			

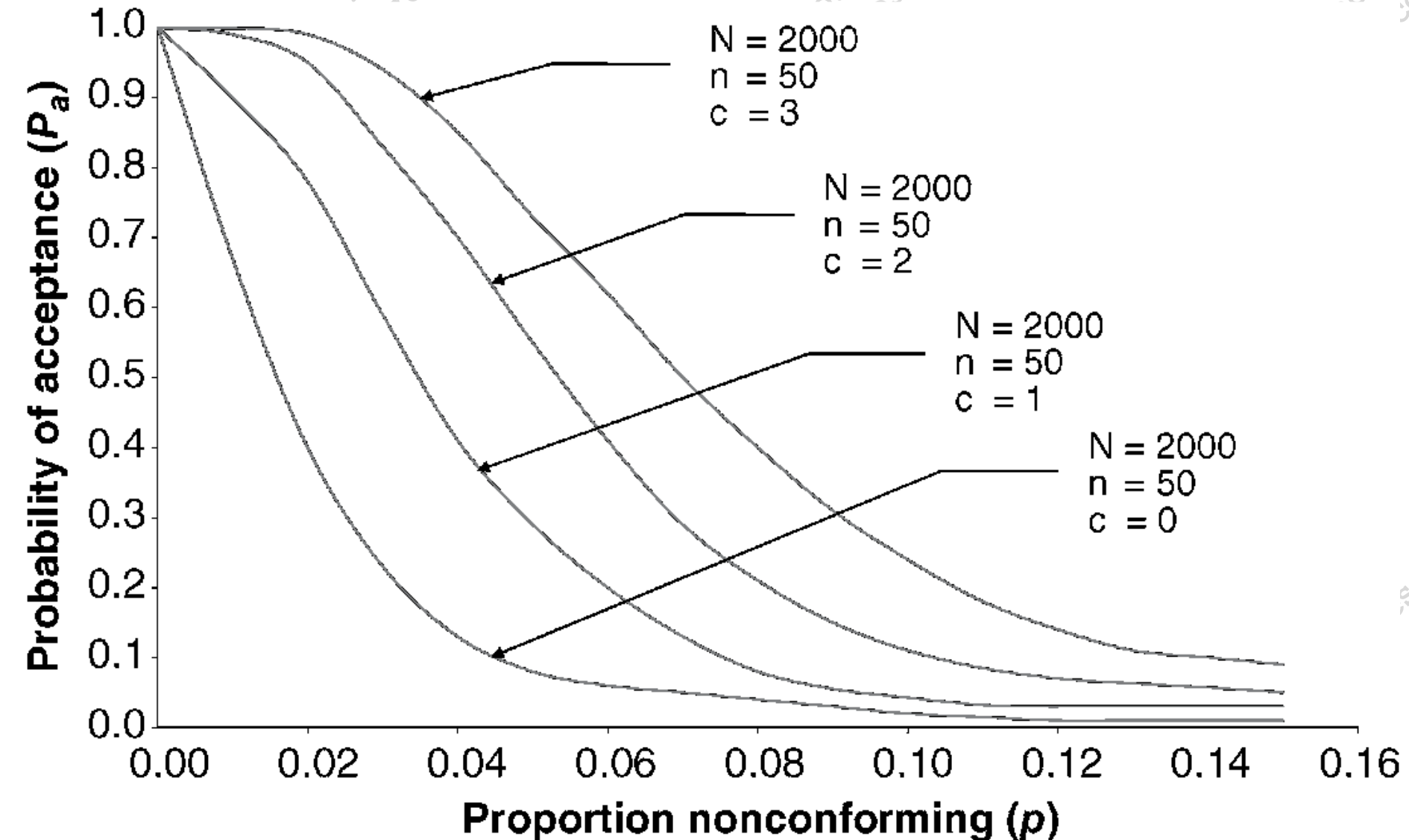
Construct an OC curve for a single sampling plan where the lot size is 2000, the sample size is 50, and the acceptance number is 2.



Effect of the sample size on the shape of the OC curve



Effect of the acceptance number on the shape of the OC curve



Types of Sampling

Sampling plans are strategies used in quality control to decide whether to accept or reject a lot (batch) of items based on inspection of a sample. There are several types of attribute sampling plans, each with different procedures and purposes.

- Single Sampling Plan
- Double Sampling Plan
- Multiple Sampling Plan
- Sequential Sampling Plan

Single Sampling Plan

- **Process:** Take one random sample of size n from the lot. Count the number of nonconforming items.
 - If the count is less than or equal to the acceptance number c , accept the lot.
 - If the count exceeds c , reject the lot.
- **Advantages:** Simple to administer and requires relatively straightforward calculation.
- **Disadvantages:** May require more inspected items on average compared to other plans, especially when the lots are of consistently good quality.

Double Sampling Plan

- **Process:** Take a first sample n_1 , then:
 - If the number of nonconforming items is $\leq c_1$, accept the lot.
 - If it is $\geq r_1$, reject the lot.
 - If it's between these values, take a second sample of size n_2 and combine findings:
 - If total nonconforming items from both samples $\leq c_2$, accept the lot.
 - If $\geq r_2$, reject the lot.
- **Advantages:** Often results in fewer inspections, on average, especially when product quality is clearly good or bad in the first sample.
- **Disadvantages:** More complex to execute and administer than single plans.

Multiple Sampling Plan

- **Process:** Similar to double sampling, but involves several possible samples and decision stages.
 - At each stage, after inspecting the added sample, you decide whether to accept, reject, or continue sampling.
- **Advantages:**
 - Tends to minimize the average number of items inspected since decisions are made as soon as possible.
- **Disadvantages:**
 - Most complex administratively; highest record-keeping and operational requirements.

Sequential Sampling Plan

- **Process:** Inspect items one at a time and after each, decide to accept, reject, or inspect more.
 - Uses a graph (acceptance and rejection lines) plotting cumulative nonconforming items against items inspected to decide when to stop.
- **Advantages:**
 - Can minimize inspection efforts—potentially leading to very few items needing inspection when quality level is clearly good or bad.
- **Disadvantages:**
 - Most mathematically complex, and requires constant record-keeping during inspection.

Comparison Table

Sampling Plan	Simplicity	Average Items Inspected	Administrative Cost	Typical Use Cases
Single	Easiest	Highest	Lowest	Simple, small, or infrequently inspected lots
Double	Moderate	Lower than single	Medium	When lot qualities vary or inspection is expensive
Multiple	Harder	Lowest	Highest	High-volume production, cost-sensitive industries
Sequential	Most complex	Potentially lowest	High	Destructive testing, urgent decisions

Evaluating Sampling Plans

Evaluating sampling plans involves analyzing how well a given plan performs, especially in the context of quality control. Several key measures are used to assess the effectiveness and efficiency of sampling plans.

Operating Characteristic (OC) Curve

- **Definition:** The OC curve plots the probability of accepting a lot (P_a) versus the proportion nonconforming in the lot (p).
- **Purpose:** It visually shows the discriminatory power of the plan—how well it can distinguish between good and poor quality lots.
- Steeper OC curves (more “S-shaped”) indicate greater discriminatory power.
- A plan should ideally accept lots with low nonconformance most of the time and rarely accept lots with high nonconformance